

IT 584 : Approximation Algorithms

Course Placement : Approximation Algorithms is an elective course for the first year (2nd Semester) students of the M.Tech program with software systems specialization. It is also open as an elective to B.Tech and B.Tech MnC students (recommended for those interested in theoretical analysis, mathematical and algorithmic aspects of CS).

Course Format: It is 3 hours of lecture and 1 hour of tutorial every week.

Credit Structure: 3-1-0-4 (L-T-P-C)

Course Content: Many problems in computer science are known to be “hard”. It is computationally very expensive to solve these exactly. The aim of this course is to learn various techniques to solve such problems in an approximate manner. The course will discuss algorithmic techniques like greedy algorithms, local search, approximation schemes, LP based algorithms, randomized algorithms and a few others. For each technique the course will cover one or more problems and also focus on detailed analysis of the approximation obtained.

Prerequisites: A course on basic algorithms and data structures, discrete mathematics. Knowledge of probability and random variables, linear algebra and optimization will be useful. Mathematical maturity and interest will help appreciate and profit from the course.

Reference Books:

There is no prescribed textbook. We will follow material from different sources. Few good reference books are:

- **Approximation Algorithms by V. Vazirani**
- **The Design of Approximation Algorithms by David Williamson and David Shmoys**
- **Randomized Algorithms by Rajeev Motwani**
- **Probability and Computing Mitzenmacher and Upfal**

Assessment Method : (Subject to modifications)

Term Paper/Flip class - 30%

Mid Sem- 30%

End Sem - 40%

Course Outcomes:

After successful completion of this course, students will be able to:

- Understand the notion of hard problems and need of approximation algorithms
- Understand techniques to design approximation algorithms for various problems
- Analyze approximation algorithms for different problems

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X		X						X	X	X

List of Topics:

Sr. No.	Description
1	Why approximation algorithms and What are approximation algorithms:NP complete problems
2	Greedy algorithms and Local Search algorithms
3	Approximation Schemes
4	Randomized Approximation Algorithms
5	Primal Dual Methods
6	Approximation algorithms for streaming setting
7	Approximation algorithms in Machine Learning